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MIL-L-70846 (AR)
14 February 1992

MILITARY SPECIFICATION

LASER TUBE/MODULATOR ASSEMBLIES, SINGLE/DUAL POWER, GENERAL SPECIFICATION FOR

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture, inspection, packaging and acceptance of the following Single/Dual Power Laser Tube/Modulator Assemblies: Dragon, 11749646; TOW, 11749647; M16A1, 11835688; M60, 11835689; M2, 11835690; M85, 11835691; VIPER, 11835694, and Controller Gun, 11835695.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1265

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11835691	-	Laser Tube/Modulator Assy, M85
11835694	-	Laser Tube/Modulator Assy, VIPER
11835695	-	Laser Tube/Modulator Assy, Controller Gun

INSPECTION EQUIPMENT (CAGE CODE 19200)

11748854	-	Detector Module, Si-PN (see 6.8)
11749020	-	Printed Wiring Board Assy, Helmet (MWLD) (see 6.8)
11749167	-	Electronics Assy, Modulator (see 6.8)
11748824	-	Laser Driver Monochip (see 6.8)

PACKAGING DATA SHEET (CAGE CODE 19200)

SPI-P11749646	-	Laser Tube/Modulator Assy, Dragon
SPI-P11749647	-	Laser Tube/Modulator Assy, TOW
SPI-P11835688	-	Laser Tube/Modulator Assy, M16A1
SPI-P11835689	-	Laser Tube/Modulator Assy, M60
SPI-P11835690	-	Laser Tube/Modulator Assy, M2
SPI-P11835691	-	Laser Tube/Modulator Assy, M85
SPI-P11835694	-	Laser Tube/Modulator Assy, VIPER
SPI-P11835695	-	Laser Tube/Modulator Assy, Controller Gun

(Copies of other Government documents, drawings and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item description. This laser tube/modulator assembly includes a laser tube assembly and a modulator PWBA. The modulator PWB contains a modulator transformer, one or two switch transistors and protective breakdown diodes, one or two current sense resistors and a temperature compensation network consisting of two thermistors and four fixed resistors.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel Manufacture and Inspection of, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-W-63150	- Weapons and Support Materiel, Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-810	- Environmental Test Methods
MIL-STD-1686	- Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices)
MIL-STD-1695	- Environments, Working, Minimum Standards for
MIL-STD-2000	- Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-45662	- Calibration System's Requirements

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HANDBOOK

MILITARY

DoD-HBK-263 - Electrostatic Discharge Control Handbook for Protection of Electrical Electronic Parts, Assemblies and Equipment (Excluding electrically initiated explosive devices) METRIC

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER.

PRODUCT DRAWINGS (CAGE CODE 19200)

9358638	-	Laser Tube Assy
11749048	-	Laser Tube Assembly
11749078	-	Printed Wiring Board Assy, Modulator
PL11749078-1	-	Printed Wiring Board Assy, Modulator, M16
PL11749078-2	-	Printed Wiring Board Assembly, Modulator, M2, M85, M60, VIPER
PL11749078-3	-	Printed Wiring Board Assembly, Modulator, Controller Gun
PL11749078-4	-	Printed Wiring Board Assembly, Modulator, Dragon
PL11749078-5	-	Printed Wiring Board Assembly, Modulator, TOW
11749204	-	Schematic Diagram, Modulator, Dual Power
11749205	-	Schematic Diagram, Laser
11749370	-	Schematic Diagram, Modulator, Single Power
11749646	-	Laser Tube/Modulator Assy, Dragon
11749647	-	Laser Tube/Modulator Assy, TOW
11835688	-	Laser Tube/Modulator Assy, M16A1
11835689	-	Laser Tube/Modulator Assy, M60
11835690	-	Laser Tube/Modulator Assy, M2

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The remotely located laser driver chip supplies pulses to turn on the transistor switch(es), one switch being used for setting the kill beam energy while the second, if used, sets the near miss energy. The primary current is monitored by means of the voltage drop across the current sense resistor(s) and a voltage comparator in the remote laser driver chip. A successful comparison with the chip's internal reference voltage results in turning off the transistor switch.

At this point, the energy stored in the transformer's magnetic field is transferred to the energy storage capacitor located in the laser tube assembly.

A subsequent fire pulse to the gate of the semiconductor controlled rectifier (SCR) (also in the laser tube assembly) then discharges the capacitor through the laser diode producing a predetermined current pulse and thereby an optical output pulse of predetermined energy.

The laser tube and modulator assemblies are configured in accordance with Table I.

TABLE I. Laser Tube/Modulator Configurations.

System	Dual/Single Power	Modulator	Laser Tube
M16	Dual	11749078-1	11749048-1
M2	Dual	11749078-2	11749048-3
M60	Dual	11749078-2	11749048-2
M85	Dual	11749078-2	11749048-4
VIPER	Dual	11749078-2	11749048-7
Cntrl Gun	Single	11749078-3	11749048-12
DRAGON	Single	11749078-4	11749048-6
TOW	Single	11749078-5	11749048-5

3.1.1 Matched assemblies. The laser tube and modulator assemblies shall be matched in the above configuration (see Table I) by selecting R6 or R7, or if necessary both R6 and R7 (see Figures 1 and 2) such that the near miss energy and irradiation energy limits for the systems specified are met (see Table II).

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TABLE II. Resistor Selection.

System	R6 (ohms)	R7 (ohms)	Near Miss Energy	Irradiance Energy
M16	2-15	6.81-24.9	3.4.2.2	3.4.2.3
M2 M60 M85 VIPER	1.1-15	4.75-24.9	3.4.2.2	3.4.2.3
Controller Gun	-	6.19-24.9	-	3.4.2.4
DRAGON	-	2.37-24.9	-	3.4.2.4
TOW	-	2.21-24.9	-	3.4.2.4

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Environmental Stress Screen (ESS). All assemblies shall be subjected to an environmental stress screen prior to conformal coating consisting of temperature cycling and soaking between the temperatures -35°C and +85°C. After exposure to ESS, the characteristics of 3.4 shall not be degraded.

3.4 Characteristics.

3.4.1 Operating conditions. The operating conditions shall be in accordance with Table III, measured under quiescent

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conditions. For terminal and component designations called out in this specification, see drawing 11749204, Schematic Diagram, Modulator, Dual Power or drawing 11749370, Schematic Diagram, Modulator, Single Power, and drawing 11749205, Schematic Diagram, Laser.

TABLE III. Operating Conditions.

Parameter	Terminal (of PWBA)	Minimum	Nominal	Maximum	Unit
Ambient Temperature		-32	25	75	°C
Input (+) Power	1	6.8	9.0	10.0	Vdc
Return (-) Power	5				
Input Current	1		2	3 1/	mA

NOTE: 1/ During code transmission, 20mA nominal, 100mA maximum.

3.4.2 Performance. Unless otherwise specified, the laser tube/modulator assembly shall meet the requirements of this specification under the following conditions:

a.	Temperature	25°±10°C (77°±18°F)
b.	Relative humidity	50±30 %
c.	Barometric pressure	725+50, -75mm Hg (28.5+2.0, -3.0 inches Hg)
d.	Input power and loads	See Figure 1 (Dual Power) See Figure 2 (Single Power)

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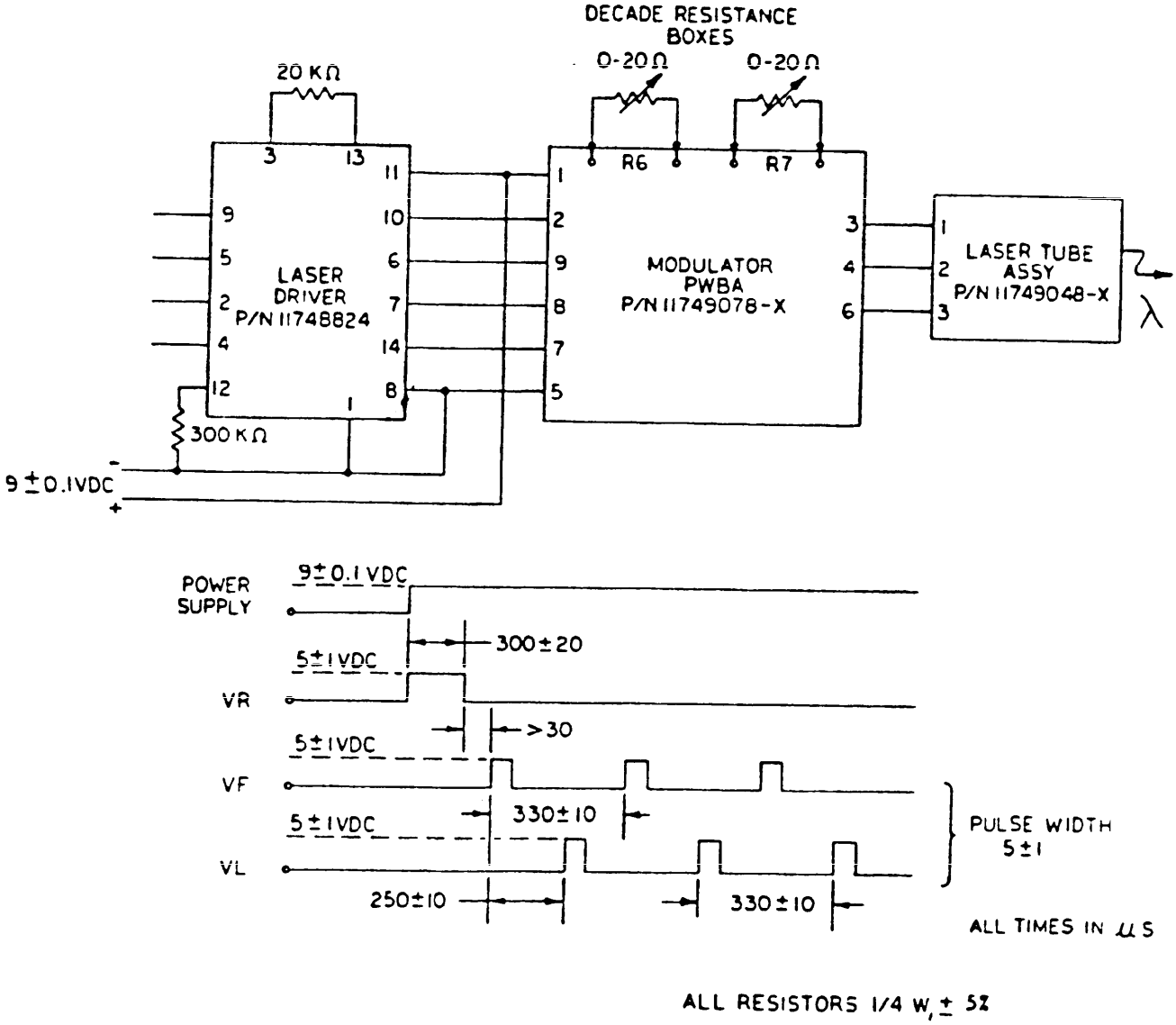


FIGURE 1. Laser Tube/Modulator PWBA Power Source and Loads. (Dual Power)

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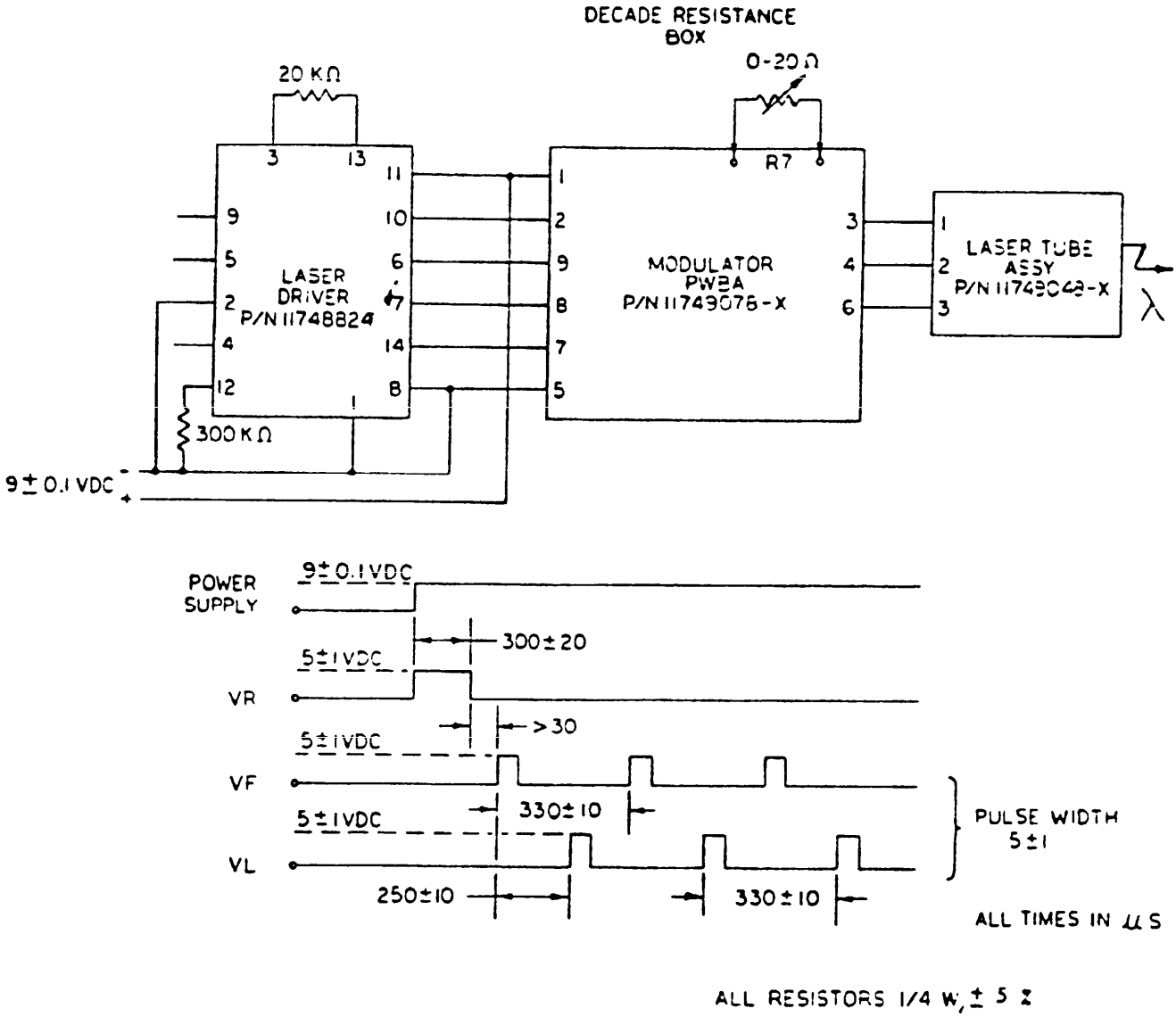


FIGURE 2. Laser Tube/Modulator PWBA Power Source and Loads. (Single Power)

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3.4.2.1 Current drain. With power applied to terminals 1 (+) and 5(-) of the Modulator, the quiescent current drain of the laser tube/modulator assembly shall be no more than 3.0mA.

3.4.2.2 Near miss energy (Dual power). With power applied to terminals 1(+) and 5(-) of the Modulator, make the following connections on the P/N 11748824 chip: apply 9 ± 0.1v to terminal 11, connect V_R to terminal 9, connect V_L to terminal 5, connect V_F to terminal 4, connect terminals 1 and 2 and the near miss energy shall be within the limits of Table IV.

TABLE IV. Laser Energy Requirements.

System Used On	Irradiance Energy Figure No. 1/	Near Miss Energy (Ergs) 2/	Max Output Energy (Ergs) 2/
Dual Power			
M16	3	0.9-2.2	N/A
M2	4	1.2-2.2	N/A
M60		1.2-2.2	N/A
M85		1.2-2.2	N/A
VIPER	6	1.2-2.2	N/A
Single Power			
Cntrl Gun	5	N/A	2.2
DRAGON	7	N/A	5.2
TOW	8	N/A	5.2

NOTES:

1/ Figures 3 through 8 are located after 3.4.2.5.

2/ 1 erg = 10^{-7} Joules

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3.4.2.3 Irradiance energy - 50 meter (Dual power). With power applied to terminals 1(+) and 5(-) of the modulator, make the following connections on the P/N 11748824 chip: apply 9± 0.1v to terminal 11, connect V_R to terminal 9, connect V_L to terminal 2, connect V_F to terminal 4, connect terminals 1 and 5 and the laser energy shall be within the values for irradiance energy at room temperature referenced in Table IV.

3.4.2.4 Irradiance energy - 50 meter (Single power). With power applied to terminals 1 (+) and 5 (-) of the Modulator, make the following connections on the P/N 11748824 chip: apply 9 ± 0.1v to terminal 11, connect V_R to terminal 9, connect V_L to terminal 5, connect V_F to terminal 4, connect terminals 1 and 2, and the laser energy shall be within the values for irradiance energy at room temperature referenced in Table IV. The maximum output energy levels are listed in Table IV.

3.4.2.5 Laser pulses. With power applied to terminals 1 (+) and 5 (-) of the modulator, make the following connections on the P/N 11748824 chip: apply 9 ± 0.1v to terminal 11, connect V_R to terminal 9, connect V_L to terminal 2 (Dual) or connect V_L to terminal 5 (Single), connect terminals 1 and 5 (Dual) or connect terminals 1 and 2 (Single), and connect V_F to terminal 4. The laser tube shall then output pulses that correspond to the V_F pulses in frequency, width, and timing.

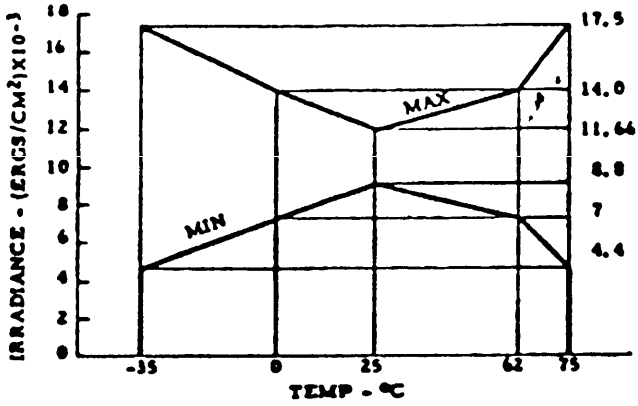


FIGURE 3. 50 Meter Beam Center Irradiance (M16).

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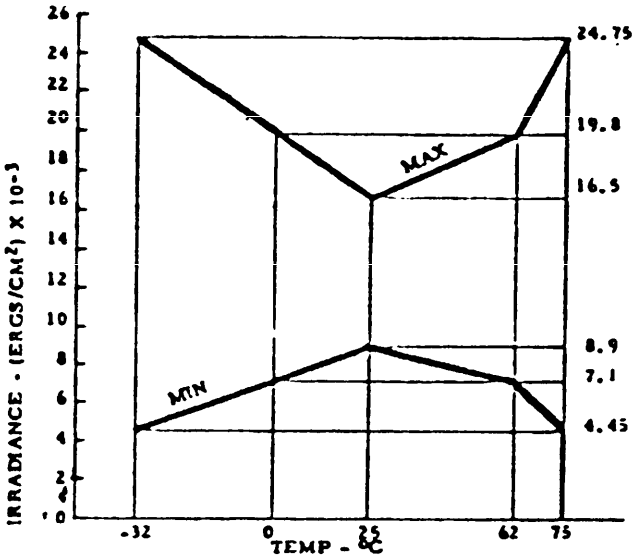


FIGURE 4. 50 Meter Beam Center Irradiance (M2, M60, M85).

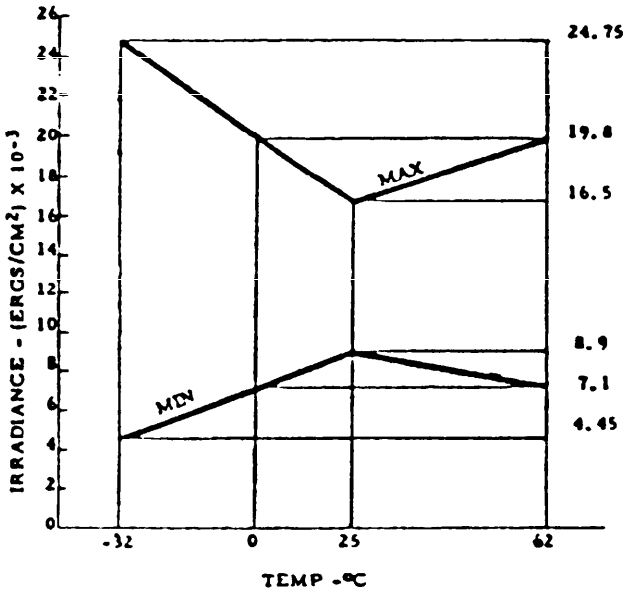


FIGURE 5. 50 Meter Beam Center Irradiance (Controller Gun).

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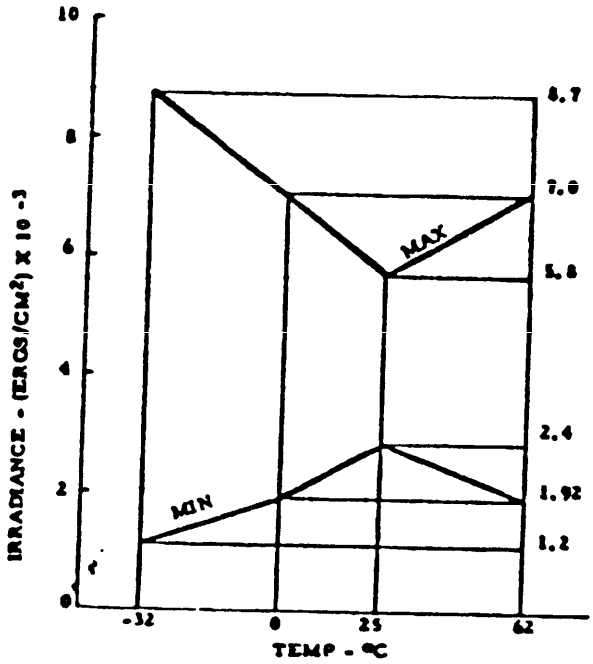


FIGURE 6. 50 Meter Beam Center Irradiance (VIPER).

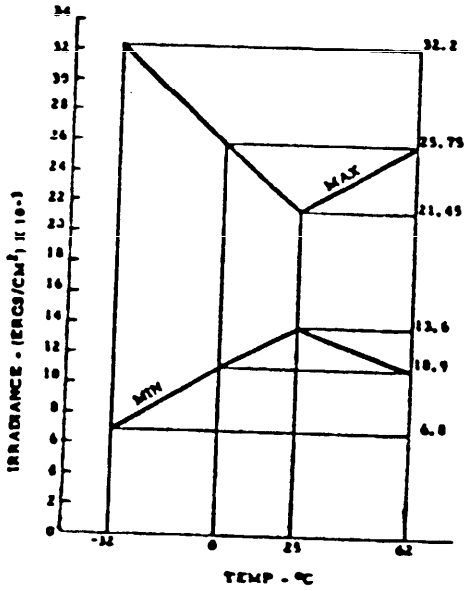


FIGURE 7. 50 Meter Beam Center Irradiance (DRAGON).

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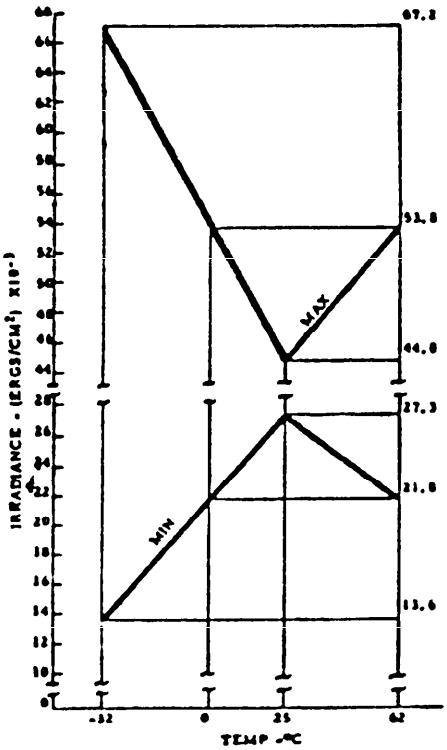


FIGURE 8. 50 Meter Beam Center Irradiance (TOW).

3.4.3 Environmental requirements. Unless otherwise specified the assembly shall be subjected to the following environmental conditions and meet the stated requirements.

3.4.3.1 High temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 when exposed to and after the following.

3.4.3.1.1 Operating temperature. The assembly shall operate up to a temperature of +75°C (+167°F).

3.4.3.1.2 Storage temperature. The assembly shall operate after exposure to a temperature of +85°C (+185°F) for a period of 48 hours.

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3.4.3.2 Low temperature. The assembly shall not be damaged and shall meet the performance requirements of 3.4.2 when exposed to and after the following.

3.4.3.2.1 Operating temperature. The assembly shall operate down to a temperature of -31.7°C (-25°F) after exposure at -31.7°C for 4 hours.

3.4.3.2.2 Storage temperature. The assembly shall operate after exposure to a temperature of -34.4°C (-30°F) for a period of 24 hours.

3.4.3.3 Shock. The assembly shall meet the performance requirements of 3.4.2 after shocks with a nominal level of 40 g's terminal peak and nominal duration of 11 milliseconds as shown in Figure 9. Apply three shocks (18 total) in each direction of the three mutually perpendicular axes.

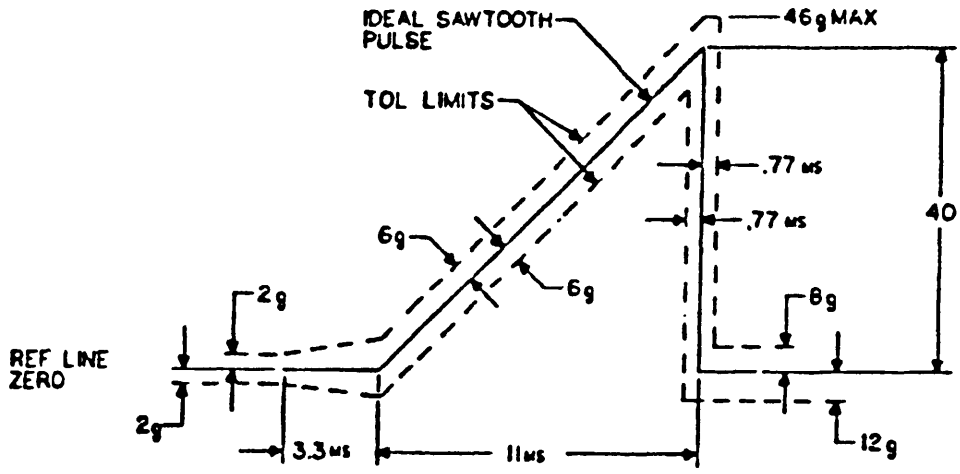


FIGURE 9. Shock Pulse.

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3.4.3.4 Vibration. The assembly shall meet the performance requirements of 3.4.2 after sinusoidal vibrations of frequency and amplitude given below and in Figure 10 on three mutually perpendicular axes. Total duration shall be 60 minutes per axis.

Frequency	Amplitude	Sweep rate
7-30 Hz	±1.5 g	7-500-7 Hz
30-50 Hz	0.033" DA	in 15 minutes
50-500 Hz	±4.2 g	

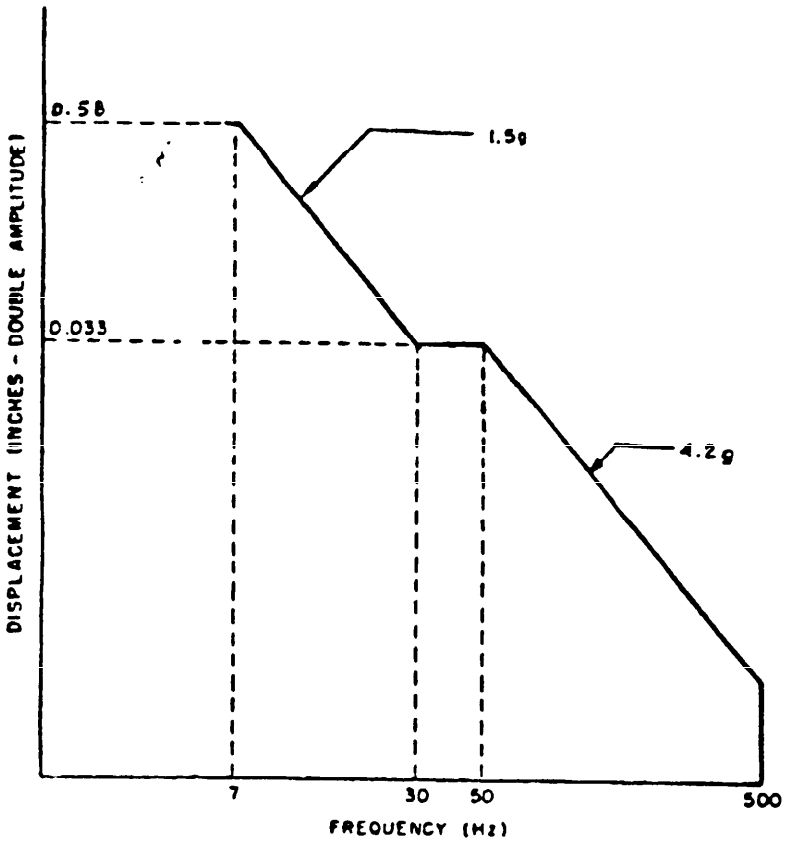


FIGURE 10. Vibration Test.

3.5 Design and construction. The PWBA and laser tube assembly shall be fabricated and assembled in accordance with drawings, parts lists, and other documents specified on drawings

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11749078 and 11749048. The PWBA and laser tube assembly shall be manufactured and tested in a working environment in accordance with Table I, work area 5 of MIL-STD-1695.

3.5.1 Standards of manufacture. Handling of Electrostatic Sensitive Parts. This equipment contains semiconductor devices such as CMOS, mosfets, and microelectronic devices which may be degraded or damaged by an inadvertent discharge of static electricity. To prevent damage to these components, the supplier shall provide for adequate handling procedures and equipment in accordance with MIL-STD-1686 and DoD-HBK-263.

3.5.2 Soldering. Soldering shall be in accordance with the general requirements of section 4 of MIL-STD-2000.

3.5.3 Workmanship. General workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Environmental stress screen (see 4.5).

c. Quality conformance inspection (see 4.6).

d. Packaging inspection (see 4.6.4).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the working environment specified in 3.5 and at standard ambient conditions defined in 3.4.2 using the methods of 4.8 and 4.9.

4.4 First article inspection.

4.4.1 Sample. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of three printed wiring board assemblies.

4.4.2 Inspections to be performed. First article inspections are listed in tables V and VI. Each first article sample unit shall be inspected in accordance with the Table V before being subjected to the environmental tests in Table VI in the sequence listed. As determined by the Government, the first article assemblies may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

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TABLE V. Examinations and Performance Tests. 1/ 2/

Class	Characteristics	Requirements Paragraph	Test/ inspection Method	First Article Sample Size	Quality 3/ Conformance Inspection Sample Size/ Acceptance Criteria
101	Current Drain	3.4.2.1	4.8.2.1	3	32/Acc-0,Rej-1
102	Near Miss Energy (Dual Power)	3.4.2.2	4.8.2.2	3	32/Acc-0,Rej-1
103	Irradiance Energy 50-meter (Dual Power)	3.4.2.3	4.8.2.3	3	32/Acc-0,Rej-1
104	Irradiance Energy 50-meter (Single Power)	3.4.2.4	4.8.2.4	3	32/Acc-0,Rej-1
105	Laser Pulses	3.4.2.5	4.8.2.5	3	32/Acc-0,Rej-1
201	Workmanship	3.5.3	Visual	3	125/Acc-0,Rej-1
202	Soldering 4/	3.5.2	Visual	3	125/Acc-0,Rej-1
203	Conformal Coating	3.5	Visual	3	125/Acc-0,Rej-1
204	Dim Characteristics	3.5	SME 5/	3	125/Acc-0,Rej-1
205	Comp Orientation	3.5	Visual	3	125/Acc-0,Rej-1
206	Comp and Board Identification	3.5	Visual	3	125/Acc-0,Rej-1

NOTES:
1/ ESS testing per 4.5 is required for all deliverable units.
2/ 200 class (minor characteristics) examinations required prior to 100 class (major characteristics) performance tests.
3/ For lot sizes less than 32 or 125 as listed, 100% inspection is required (see 4.6.5).
4/ Check before conformal coating is applied.
5/ Standard Measuring Equipment.

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4.4.3 Rejection. If any assembly fails to comply with Tables V and VI any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly to comply with any of the requirements.

4.5 Environmental stress screen. An environmental stress screening program shall be established. All units shall be subjected to a thermal stress screen in accordance with Table VII and the procedure below.

4.5.1. Conformal coating. Thermal stress screen shall be performed prior to conformal coating unless another method is approved by the Government.

TABLE VI. Environmental Tests.

Class	Characteristic	Requirements Paragraph	Test Paragraph	First Article Sample Size	Quality Conformance Inspection Sample Size Acceptance Criteria
106	High Temp	3.4.3.1	4.9.1	3	1/Acc-0, Rej-1
107	Low Temp	3.4.3.2	4.9.2	3	1/Acc-0, Rej-1
108	Shock	3.4.3.3	4.9.3	3	1/
109	Vibration	3.4.3.4	4.9.4	3	1/

NOTE:
1/ Not required for quality conformance inspection.

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TABLE VII. ESS Temperature Cycling Requirements.

Low Temp	High Temp	Rate Change	Power	Monitor	Cycles/ Group A Test
-34.4°C	+85°C	15°C/minute	Off	No	20/Failure Free or Additional Cycle

4.5.2 Chamber temperature. Chamber temperature cycling shall be from -34.4°C to +85°C (-30°F to +185°F) and the duration of one cycle to be 8 minutes up and 8 minutes down plus 4 minutes dwell at high temperature and 4 minutes at low temperature for a total cycle duration of 24 minutes. The units shall begin and end the cycles at ambient temperature (see Figure 11).

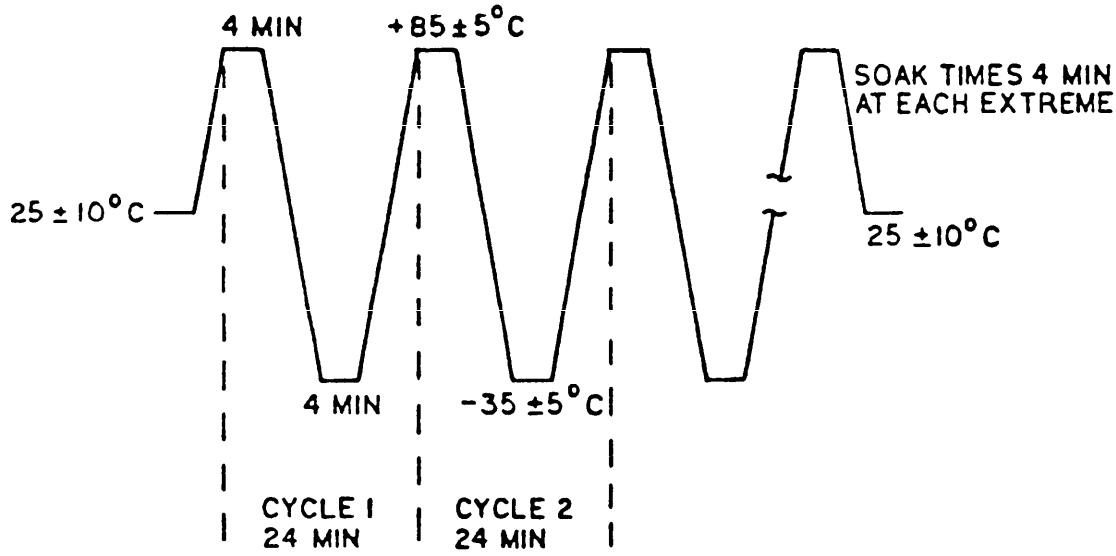


FIGURE 11. Environmental Stress Screen Temperature Profile.

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4.5.3 Cycle frequency. A total of twenty temperature cycles are to be imposed on each unit.

4.5.4 Temperature cycles. Temperature cycling shall be conducted with units nonoperating and without power (static).

4.5.5 Test requirements. After the thermal stress screen, the units shall meet the Group A tests (4.6.2). If a unit is rejected, the unit shall be repaired for the failure (if possible) and be put through one additional cycle of ESS. The unit shall then undergo and pass the Group A tests.

4.6 Quality conformance inspection. Quality conformance inspection requirements are defined in 4.6.2 and 4.6.3.

4.6.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn of which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior of subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.6.2 Group A tests. Group A tests consist of examinations and performance tests listed in Table V in the quantities specified.

4.6.3 Group B tests. Group B tests shall consist of the tests indicated in Table VI and shall be conducted on one sample unit per inspection lot which has been subjected to, and passed, Group A tests (see 4.6.2).

4.6.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall verify that the units are packaged in accordance with the requirements

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of 5.1. During the period of first article inspection, one additional unit, packaged in accordance with 5.1, shall be examined to the requirements for quality conformance inspection listed below. Failure of packaging shall be cause for rejection of packaging. Two units per lot shall be examined for the following:

a. Verify that the outside bag is marked in accordance with MIL-STD-129 and is properly closed.

b. Verify that sleeve is around box and that foam properly surrounds inner bag.

c. Verify that the inner bag is properly sealed.

d. Verify that the unit is not damaged after the removal of all packaging materials.

4.6.5 Failure of sample. Should any one sample fail to meet any of the specified examination and test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract (see 6.7).

4.7 Inspection equipment.

4.7.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607 and MIL-W-63150.

4.7.2 Contractor-furnished inspection equipment.

4.7.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAP/QARs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs (see 6.3).

4.7.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the Methods of Inspection specified in 4.8 and 4.9 of this specification and with the component inspection procedures specified in Examination and Test Facilities requirements of MIL-F-13926, whenever

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Government designs are not specified (see 4.7.2.1). Since tolerance of test equipment is normally considered to be within ten percent of product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. When specified in contract or purchase order, the contractor shall submit inspection equipment designs for Government review and evaluation (see 6.3).

4.7.2.3 Test equipment. In conjunction with 4.7.2.1 and 4.7.2.2, the test equipment listed in Table VIII (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.8.

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TABLE VIII. Test Equipment List.1/

Power Supplies (DC)	Standard - capable of providing levels and limits stated in 4.8.1 and 4.8.2.
Ammeter/Multimeter	Standard - capable of measuring currents stated in 4.8.2.
Pulse Generator	Standard - capable of providing pulses needed in 4.8.1 and 4.8.2.
Resistance Decade Boxes	Standard - capable of providing ohmage stated in 4.8.1 and 4.8.2.
Laser Energy Meter and Pulse Integrator	Equivalent to EG & G Model 460-1A and 460-3 - capable of measuring laser energy specified in 4.8.1 and 4.8.2.
Laser Detector Head	Equivalent to EG & G Model 460-2 - capable of detecting laser energy specified in 4.8.1 and 4.8.2.
Radiometer/Photometer and Pulse Integrator	Equivalent to EG & G 550-1 and 550-3 - capable of measuring irradiance laser energy specified in 4.8.1 and 4.8.2
Silicon Multiprobe	Equivalent to EG & G 550-2 - capable of detecting irradiance laser energy specified in 4.8.1 and 4.8.2.
Laser Driver Monochip	P/N 11748824 - used in 4.8.1. and 4.8.2.
Si-PN Detector Module	P/N 11748854 - used in 4.8.2.5.
Helmet (MWLD) PWBA	P/N 11749020 - used in 4.8.2.5.
Dual Channel Oscilloscope	Standard - capable of measuring pulses in 4.8.2.5.

NOTE:
1/ Must be calibrated to the requirements of MIL-STD-45662 prior

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4.8 Methods of inspection. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the Government procuring agency for approval prior to use (see 4.7.2.2).

4.8.1 Matched assemblies. To obtain the matched assemblies, connect each laser tube/modulator assembly into the circuit of Figure 12 (Dual Power) or Figure 13 (Single Power) for the purpose of identifying resistor values R6 and R7 of the assemblies. In order to meet the requirements of 3.1.1, apply power by turning on S1 and leaving all other switches as shown unless otherwise specified. Set the pulse generator to provide the signals V_R, V_L and V_F of Figure 1 and 2. Turn on the laser power meter, set the meter and detector head to 904nm and set the decade range of the integrator module to 10^{-6}. For the radiometer/photometer, turn on the autorange, set the unit selector thumbwheel and the detector switch to J/cm^2 and set the decade range on the integrator module to 10^{-3}. (Note that 1 erg = 10^{-7} Joules).

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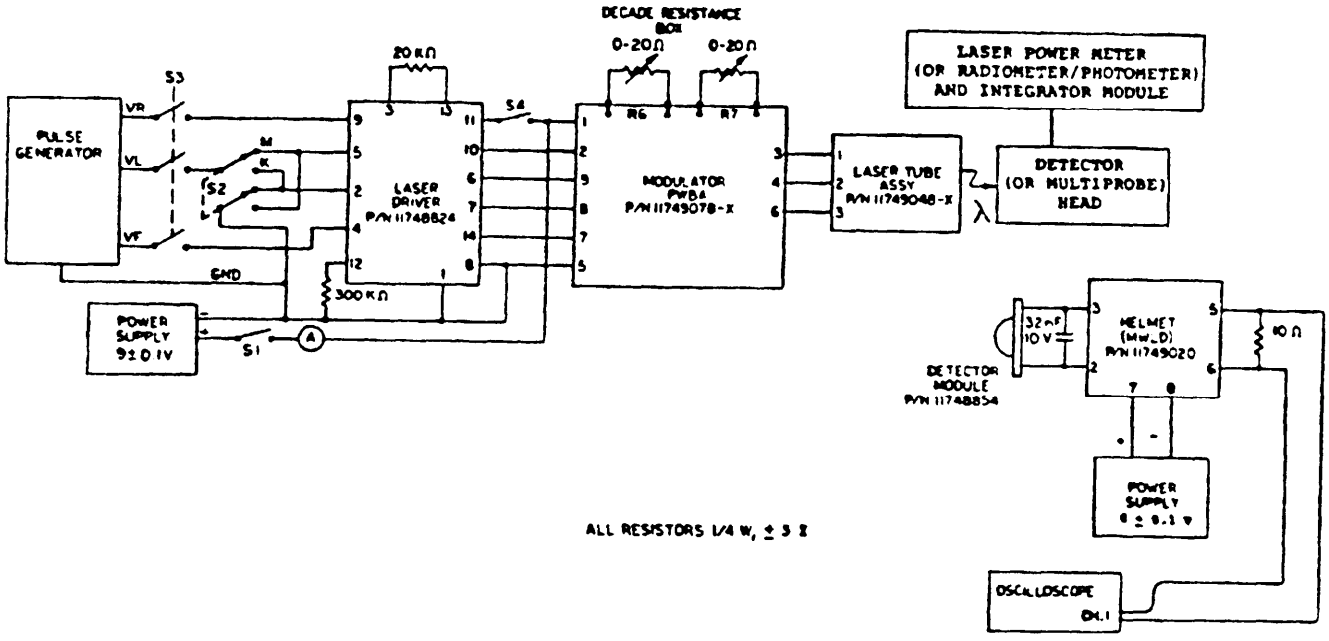


FIGURE 12. Test Circuit for Modulator (Dual Power).

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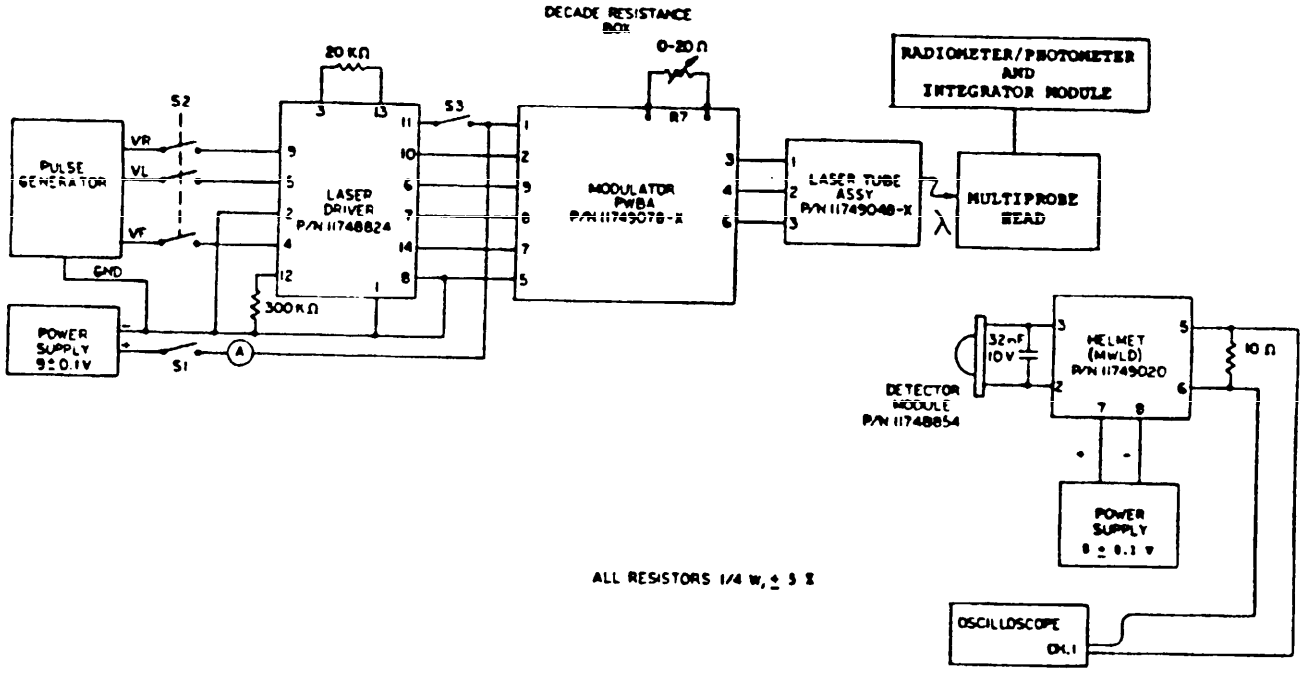


FIGURE 13. Test Circuit for Modulator (Single Power).

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4.8.1.1 Setting R6. To set R6 for the dual power assemblies, set R7 to a nominal value for each system from the range given in 3.1.1, set S2 to position M, turn ON S3 and S4, align the power meter and energy detector with the laser and adjust R6 so that the appropriate near miss energy of 3.1.1 is indicated on the power meter. Replace R6 with a fixed resistor given in PL11749078 nearest in value.

4.8.1.2 Setting R7. To set R7 for the dual power assemblies, turn ON S3 and S4, set S2 to position K, align the radiometer and multiprobe with the laser and adjust R7 so that the appropriate irradiance energy of 3.1.1 is indicated on the radiometer. For the single power assemblies, turn ON S2 and S3, align the radiometer and multiprobe with the laser and adjust R7 so that the appropriate irradiance energy of 3.1.1 is obtained on the radiometer. Replace R7 with a fixed resistor given in PL11749078 nearest in value.

4.8.2 Performance tests. After R6 and R7 has been fixed the following tests and examinations shall be performed to demonstrate compliance with 3.4.2. The PWBA shall be connected to the circuit specified in Figure 12 (Dual Power) and Figure 13 (Single Power). Apply power by turning ON S1 and leaving on until all tests have been completed. All other switches shall remain as shown unless otherwise specified in 4.8.2.1 through 4.8.2.5. Turn on the laser power meter, set the detector head and attenuator to 904 nm and set the decade range of the integrator module to 10^{-6}. For the radiometer/photometer, turn on the autorange, set the unit selector thumbwheel and the detector switch to J/cm^2 and set the decade range on the integrator module to 10^{-3} (Note that 1 erg = 10^{-7} Joules). Set the channels of the pulse generator to provide the pulses shown in Figures 1 and 2 for signals V_R, V_L and V_F.

4.8.2.1 Current drain. Verify that the quiescent current drain meets the requirements of 3.4.2.1.

4.8.2.2 Near miss energy (Dual power). To verify the dual power near miss energy requirements of 3.4.2.2, proceed as follows: turn ON S3 and S4, set S2 to position M, align the laser detector head with the laser and verify the energy ranges given in Table IV are obtained by the power meter.

4.8.2.3 Irradiance energy - 50 meter (Dual power). To verify the dual power irradiance energy requirements of 3.4.2.3,

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proceed as follows: turn ON S3 and S4, set S2 to position K, align the silicon multiprobe with the laser and verify the energy density limits in Table IV are obtained by the radiometer.

4.8.2.4 Irradiance energy - 50 meter (Single power). To verify the single power irradiance energy requirements of 3.4.2.4, proceed as follows: turn ON S2 and S3, align the silicon multiprobe with the laser and verify the energy density limits in Table IV are obtained by the radiometer. Align the laser detector head with the laser and verify the maximum output energy read on the power meter is not exceeded.

4.8.2.5 Laser pulses. Test the laser pulse by replacing the radiometer and power meter with the detector module and helmet (MWLD) combination. Power the helmet (MWLD) and the oscilloscope, turn ON S3 (single and dual power), turn ON S4 (dual power), turn ON S2 (single power) or set S2 to position K (dual power), and verify the requirements of 3.4.2.5 are met by making measurements with the oscilloscope.

4.9 Environmental tests.

a. Unless otherwise specified, environmental tests shall be conducted in accordance with the test conditions and general test performance guidance of MIL-STD-810. Test facilities, chambers and apparatus used in conducting the tests required by this specification shall be capable of meeting the conditions specified.

b. Where necessary in the following tests, it shall be permissible to temporarily open chamber doors to permit access to and manipulation of test equipment and sample.

4.9.1 High temperature. This satisfies the storage and operating conditions of high temperature requirements as specified in 3.4.3.

4.9.1.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 501 using the following procedure:

a. Step 1 - Place the test item, configured for high temperature test, into the temperature chamber.

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b. Step 2 - Raise the internal temperature of 85°C (185°F).

c. Step 3 - Maintain the internal chamber temperature for a period of 48 hours while insuring the relative humidity is not in excess of 15 percent.

d. Step 4 - Adjust the internal chamber temperature to 75°C (167°F), the highest operating temperature for which the test item is designed to operate, and maintain until temperature stabilization of the item is reached.

e. Step 5 - Operate the test item until the item is stabilized and perform the tests required in 4.8.2.

f. Step 6 - Return the test item, nonoperating, to standard ambient condition and stabilize.

g. Step 7 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.2

4.9.2 Low temperature. This satisfies the storage and operating conditions of low temperature requirements as specified in 3.4.3.

4.9.2.1 Storage and operating. The assembly shall be tested in accordance with MIL-STD-810, Method 502 using the following procedure:

a. Step 1 - Place the test item, configured for low temperature test, into the temperature chamber.

b. Step 2 - Lower the internal chamber temperature to -34.4°C (-30°F) and maintain for a period of 24 hours after stabilization.

c. Step 3 - Adjust the internal chamber temperature to -31.7°C (-25°F) and maintain the temperature for a period of 4 hours.

d. Step 4 - Operate the test item until the item is stabilized and perform the tests required in 4.8.2.

e. Step 5 - Return the test item, nonoperating, to

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standard ambient conditions and stabilize.

f. Step 6 - Visually inspect for physical damage and in the absence of physical damage, perform the tests required in 4.8.2.

4.9.3 Shock. The assembly, when mounted in the Modulator Electronics Assembly (P/N 11749167), its next higher assembly, shall be tested in accordance with MIL-STD-810, Method 516.3, Section II, Procedure I, with the following modifications:

a. Step 3 - the assembly shall be in a non-operational mode.

b. The step 3 test transient shall be in accordance with 3.4.3.3.

c. Step 5 - the assembly inspection and functional verifications shall not be conducted until shock has been completed in all three axes.

d. After shock testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.2.

4.9.4 Vibration. The assembly, when mounted in the Modulator Electronics Assembly, (P/N 11749167), its next higher assembly, shall be tested in accordance with MIL-STD-810, Method 514.3, Section II, Procedure I, for a duration of 60 minutes with the following modifications:

a. Step 4 - mount the test item on vibration equipment.

b. Step 5 - expose the assembly to test level and durations as specified in 3.4.3.4.

c. Step 6 - does not apply.

d. Step 7 - the assembly inspection and functional verification shall not be conducted until vibration has been completed in all three axes.

e. After vibration testing the assembly shall not be damaged and shall be subjected to the performance tests specified in 4.8.2.

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4.10 Certifiations. Certification shall be provided that all units have been produced in accordance with the requirements of 3.5 and packaged in accordance with the requirements of 5.1.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with the Special Packaging Instructions: P11749646, P11749647, P11835688, P11835689, P11835690, P11835691, P11835694, and P11835695. The level of protection shall be A, B, or C, as defined by the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The following Single/Dual Power laser tube/modulator assemblies are a part of the Multiple Integrated Laser Engagement System: Dragon, P/N 11749646; TOW, P/N 11749647; M16A1, P/N 11835688; M60, P/N 11835689; M2, P/N 11835690; M85, P/N 11835691; Viper, P/N 11835694; Controller Gun, P/N 11835695.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number and date of this specification (see page 1).

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Special tests and examinations required (see 4.2); the type of inspection required; and the activity responsible for testing, and its facility (see 4.1).

d. Whether a first article sample is required (see 3.2), and if so, pertinent details (see 4.4) including disposition of the first article sample.

e. Certification of conformance for each lot or shipment of product.

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f. Whether Government-Loaned Property (Government Furnished inspection/test equipment) of 6.8 is available or will the contractor be required to furnish inspection equipment (see 4.7).

g. Applicable level(s) of preservation, packaging, packing and marking instructions (see 5.1).

6.3 Submission of contractor inspection equipment designs for approval. Unless otherwise specified, submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423, in the contract. The letter of submittal shall specify, the contractor; contract number; title, number and revision of this specification; item nomenclature; classification of defect and test paragraph.

6.4 Definitions.

6.4.1 MILES. Multiple Integrated Laser Engagement System.

6.4.2 SME. Standard Measuring Equipment.

6.5 Drawings. Drawings listed in Section 2 of this specification under heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group

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A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

6.7 Government-loaned property. Unless otherwise specified (see 6.2), the contracting officer should arrange to loan the following Government-furnished Inspection/Test equipment specified below:

P/N	Title	Ref. Paragraph
11748854	Detector, Module Si-PN	4.8.2.5
11749020	PWBA, Helmet (MWLD)	4.8.2.5
11749167	Electronics Assy, Modulator	4.9.3,4.9.4
11748824	Laser Driver Monochip	4.8.2

6.8 Subject term (key word) listing.

Fire Control
Training Devices

Custodians:
Army - AR

Preparing activity
Army - AR

(Project 1265-A010)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-70846 (AR)	14 FEBRUARY 1992

3. DOCUMENT TITLE
LASER TUBE/MODULATOR ASSEMBLIES, SINGLE/DUAL POWER, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
DDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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